STATISTICS

Maximum Marks: 100 **(Theory: 70: Practical: 30)**

Unit I: Introduction to Statistics **(06 marks)**

Historical overview of Statistics, Definition and Meaning of Statistics, Importance and Scope of Statistics, Statistics Then and Now, New Career options in Statistics, Role of Statistics in Integrated research. Limitations of Statistics.

Unit II: Statistical Data Collection **(06 marks)**

Data and its type (Primary, Secondary, Qualitative and Quantitative data), Sources of Secondary data. Method of data collection (Questionnaire and Interview Method). Merits and demerits of these Methods. Presentation of data, Classification and tabulation of data. Discrete and continuous data. Frequency and frequency distribution. Concept of Population and Sample.

Unit III: Graphical Representation of Data **(07 marks)**

Representation of data by Graph/diagram, advantages of graphical representation of data, Construction of diagrams/ Charts (Bar chart, Multiple Bar diagram, Sub-divided bar chart, Pie chart), Frequency graphs (Histogram, Frequency Polygon), Cumulative frequency curves (Ogive/ Ojive).

Unit IV: Measures of Location **(08 marks)**

Central Tendency meaning, Different Measures of Central Tendency (Mean, Median, Mode, Geometric Mean and Harmonic Mean), Computation of Measures of Central Tendency from Discrete and Continuous data, Essentials of good average. Merits and Demerits of Measures of Central Tendency. Combined Mean and Weighted Mean.

Unit V: Partition Values. **(08 marks)**

Concept of Partition values, Distinguish Measure of central tendency and Measures of location, Uses of Partition values, Graphical representation of Median, Concept of Quartiles, Deciles and Percentiles. Percentile Rank, Empirical relation between Mean, Median and Mode, Symmetrical and Asymmetrical data.

Unit VI: Dispersion **(09 marks)**

Dispersion and its absolute measures (Rang, Quartile Deviation, Mean Deviation and Standard Deviation). Merits and Demerits of these measures. Relative measures of Dispersion (Co-efficient of Range, Co-efficient of Quartile deviation, Co-efficient of Standard deviation). Co-efficient of variation (C.V).

Unit VII: Moments, Skewness and Kurtosis **(10 marks)**

Define Moments Types of Moments (Raw Moments and Centre Moments for discrete and Continuous data) Relationship between Raw and Central Moments

Define Skewness and its types, Measures of Skewness (Karl Pearson, Bowleys and

Moment based measure). Kurtosis and its types. Measures of Kurtosis.

Unit VIII: Correlation **(08 marks)**

Concept of Bi-Variate data, Scattered diagram, Concept of Correlation and its types. Methods of Measuring Correlation coefficients (Product moment method, Graphical method). Properties of Correlation coefficient. Rank correlation for simple and repeated Ranks.

Unit IX: Linear Programming and Computer Applications **(08 marks)**

Basic concept of inequalities, Solution of one variable inequalities, Graphical representation of inequalities, Basic concept of Linear programming.

Introduction to Computers, uses of computers in Education, Various components /Units of Computer (Input /Output), Hardware and Software, Concept of flow charts.

Scheme of Assessment (Theory)

The Question paper shall contain - Four sections A, B, C and D. Each section is compulsory.

1. Section A-Question 1 to 10 comprises of 10 questions of 1 mark each (MCQ's, Fill in the blanks, True/False, Assertion-Reason etc.).
2. Section B-Question 11 to 19 comprises of 9 Very Short Answer (VSA)-type questions of 2 marks each.
3. Section C-Question 20 to 28 comprises of 9 Short Answer (SA)-type questions of 3 marks each.
4. Section D-Question 29 to 31 comprises of 3 Long Answer (LSA)-type questions of 5 marks each with internal choice

PATTERN	MARKS	QUESTIONS	TOTAL
Very Short Questions	01	10	10 Marks
Short Type-I Questions	02	9	18 Marks
Short Type-II Questions	03	9	27 Marks
Long Type Questions	05	3	15 Marks
TOTAL		31	70 Marks

Practical/Project work **(30 marks)**

1. Collection of different categories of data from Schools/Societies/Surrounding area of Institution.
2. Construct different statistical tables from real life data (Frequency table, cumulative frequency table, Exclusive and inclusive tables)
3. Construct diagrams/charts (Bar Charts, Multiple Bar diagram, Pie Chart) from the given data.
4. Construct frequency and cumulative frequency curves to the given data.
5. Evaluate different measures of central tendency from the real-life data.
6. Evaluate different measures of location from given data.

7. Evaluate different measures of dispersion and relative measures of dispersion.
8. Compute raw and central moments from collected data.
9. Estimate Skewness and Kurtosis from given data or Moments.
10. Estimate correlation coefficient from the Discrete and continuous data.
11. Estimate Rank correlation coefficient from ranked and unranked data.
12. Construct a Linear Programming problem from any industrial or social data.

NOTE:

1. **The internal assessment/ internal Project Work will be evaluated by the concerned subject teacher.**